2016 F=ma Exam: Problem 25

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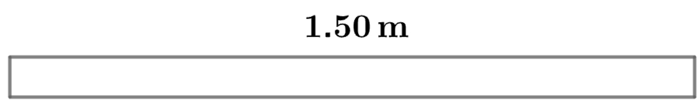

Recall independent errors add in quadrature:

$$\Delta(A+B)=\sqrt{(\Delta A)^2+(\Delta B)^2}$$

Alice: One measurement with uncertainty $\Delta L_2 = 2\,\mathrm{mm}$.

$$\Delta L_{\mathrm{Alice}}=\Delta L_2 = 2\,\mathrm{mm}$$

Bob: Two measurements with uncertainty $\Delta L_2 = 2\,\mathrm{mm}$ added together.

$$\Delta L_{\mathrm{Bob}}=\sqrt{(\Delta L_2)^2+(\Delta L_2)^2}=\sqrt{2}(\Delta L_2)=2.8\,\mathrm{mm}$$

Christina: Two measurements with uncertainty $\Delta L_1 = 1\,\mathrm{mm}$ added together.

$$\Delta L_{\mathrm{Christina}}=\sqrt{(\Delta L_1)^2+(\Delta L_1)^2}=\sqrt{2}(\Delta L_1)=1.4\,\mathrm{mm}$$

Thus,

$$\Delta L_{\mathrm{Christina}}<\Delta L_{\mathrm{Alice}}<\Delta L_{\mathrm{Bob}}$$

so the answer is $\boxed{\mathrm{A}}$.